

Steven Dyer

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EDUCATION

Memorial University of Newfoundland

Bachelor of Mechatronics Engineering; GPA: 3.8/4.0

St. John's, NL, Canada

2023 – 2028

EXPERIENCE

Kraken Robotics

Mount Pearl, NL, Canada

Embedded Engineering Intern

May. – Aug. 2026

- Developed C++ firmware for new product PCBs, enabling automated hardware validation during bring-up and accelerating SeaPower Batteries qualification timeline.
- Wrote drivers for ARM Cortex-M microcontrollers communicating over I2C, SPI, RS-485, and Ethernet, validated against hardware during iterative board spins.
- Designed PCBs for communicating with a new sonar unit through multiple interfaces, including I2C, UART, SPI, RS-485, and Ethernet, as well as PCBs for system testing during production.
- Tested PCBs using oscilloscopes, logic analyzers, and multimeters.

Formula MUN

St. John's, NL, Canada

Co-Founder & Low Voltage Lead

Apr. 2026 – Present

- Architected the vehicle's dual-CAN network, separating latency-sensitive powertrain traffic at 500 kbit/s from chassis messaging at 250 kbit/s, both controlled by an ARM Cortex-M7 microcontroller.
- Designed a PCB to successfully shut off the high voltage battery when demanding simultaneous hard throttle and braking, as well as architecting PCBs for power regulation and communication to other electronic components.
- Co-founded Memorial University's FSAE Electric team and led its low voltage team, establishing the electrical design and firmware development process, as well as their review structure.

Paradigm Engineering

St. John's, NL, Canada

Software Team Lead

Oct. 2023 – Present

- Wrote all firmware, including low-level drivers for CAN, UART, and PWM, using C on an ARM Cortex-M4 microcontroller to facilitate communication between a Jetson Orin, a Pixhawk 6C, and several motors.
- Developed a computer vision system using a ZED 2i Stereo Camera and PyTorch, implementing a CNN to detect white lines on the race track, enabling autonomous navigation and waypoint correction.
- Designed the software architecture of an autonomous go-kart which achieved second place at the 2025 Autonomous Karting Series National Grand Prix.
- Led and mentored the software team by preparing onboarding resources, documentation, and task assignments, resulting in significant contributions and experience gain by 10+ members.

Michelin

Waterville, NS, Canada

Automation Engineering Intern

Jan. – Apr. 2025, Sept. – Dec. 2025

- Developed C firmware for a new standards training lab on an ESP32 microcontroller, including I2C and UART drivers for proximity and barometric pressure sensors.
- Led a team of technicians in the installation of energy monitoring devices to improve energy tracking of machines.

PROJECTS

The World's Worst Audio Interface

May. 2026 – Present

- Designed a custom audio interface PCB in KiCAD, controlled by an ARM Cortex-M7 microcontroller, integrated with an audio codec and USB for communication to a host PC.
- Developed bare-metal C firmware on the microcontroller to emulate audio effects, such as gain, chorus, and delay.

SKILLS

Languages: C, C++, Python, JavaScript, Rust, Bash

Tools: Linux, FreeRTOS, Git, Altium Designer, KiCAD, Make, CMake, Docker, ROS2, LTSpice, GDB

Protocols: Ethernet, TCP, UDP, UART, USB, I2C, SPI, CAN, LIN, RS-485, BLE

Hardware: Digital & Analog Circuit Design, PCB Design, Oscilloscopes, Board Bring-Up, Soldering